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How Many Vowels in Proto-Coptic?

The Case for a Two-Vowel System

Marwan Kilani[1]

Abstract

This study presents a comprehensive reassessment of the Proto-Coptic vocalic system by challenging the traditional reconstruction of three phonemic vowels **a*, **i*, and **u*. I argue that this long-standing framework is methodologically problematic, as it is rooted in obsolete, early twentieth-century assumptions about a Semitic origin of Egyptian, and in a literal interpretation of Late Bronze Age cuneiform transcriptions that may not accurately reflect the phonemic reality of Egyptian.

Adopting a data-driven approach grounded in internal Coptic evidence and distributional analyses, this paper proposes an alternative model based on a binary opposition between only two proto-vowels, here represented as **a* and **E*. A central claim of this model is that the opposition between the Coptic vowels ⲓ /i/ and ⲏ /e/ does not correspond to two distinct proto-phonemes. Rather, these vowels are analysed as allophonic realizations of a single underlying vowel, whose surface quality is conditioned by its phonological environment. Specifically, the realization of this vowel appears to depend on the nature of following unstressed vowels or on the presence of specific consonants.

Although simpler than traditional reconstructions, the proposed two-vowel system yields a more coherent account of the available data. It aligns more naturally with binary morphological oppositions and offers greater explanatory power for the diachronic development of the Egyptian language.

1 Introduction

The question of the vocalization of pre-Coptic phases of the Egyptian language has attracted scholarly attention for over a century. This sustained interest has not been driven merely by a general curiosity about an otherwise unrecorded aspect of the language. Rather, it reflects the recognition that a better understanding of pre-Coptic vocalization can shed light on fundamental aspects of Egyptian grammar, and can also contribute to our understanding of the language's origins and position within Afroasiatic, its interactions with neighbouring languages (e.g. as a tool for dating loanwords), as well as a range of

1 University of Basel (marwan.kilani@unibas.ch).
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other broader sociocultural issues (e.g. questions of dialectal and sociolectal variation, and the sociocultural realities associated with them).

Yet, despite the long history of research on the topic, a closer examination of current models reveals several important inconsistencies, which signal that they still fall short of providing a fully satisfactory account. Part of the difficulty seems to lie in the way the problem has typically been approached: scholars have often relied on piecemeal solutions, introducing additional rules or variants to account for apparent exceptions, while seldom re-examining the basic assumptions of the models themselves.

This observation forms the starting point for the present paper, which aims to offer a comprehensive reassessment and a new model for the Proto-Coptic vocalization. The central innovation of this model is the proposal that Coptic vowels can be derived from only two, rather than three, proto-vowels. In addition, it will be argued that the opposition between ⲓ |i| and ⲏ |e|, which has posed the greatest challenge for current models, can be understood as an allophonic variation of a single phonemic vowel, conditioned primarily by the nature of the following unstressed vowel or by the presence of adjacent weak or weakened consonants.

While simpler than traditional three-vowel models, the two-vowel approach presented here provides a more consistent account of the data, aligns more naturally with the binary oppositions observed across various morphological domains, and opens promising avenues for further research.

2 Structure of the Paper

The paper is structured as follows. First, a series of conventions for the transcription of Egyptian and Coptic forms is presented, alongside the conceptual frames underlying the paper (§3). The core of the argument begins in §4, which deconstructs the traditional three-vowel model by examining the "Semitic bias" and the limitations of cuneiform transcriptions.

The following section (§5) presents a data-driven proposal for a two-vowel system, and argues that the reconstruction of only two phonemic vowels, **a* and **E*, is sufficient to account for the Coptic evidence. This section specifically argues that the opposition between the Coptic vowels ⲓ |i| and ⲏ |e| reflects a historically allophonic variation rather than distinct phonemes, and it proposes a set of rules to explain their distribution. Finally, the paper explores the morphological implications of this binary model and provides concluding remarks on its explanatory power for the historical development of the Egyptian language (§6).

3 Conventions, Abbreviations, and Conceptual Frames

3.1 Standard Syllabic Theory

The present paper follows the standard syllabic theory, which is based on three main tenets:

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1) Proto-Coptic permitted only two primary syllable types: open CV and closed CVC.
2) CVCVCV(C) sequences were regularly reduced to CVCCV(C) through syncope of the second vowel.
3) Stress could fall only on one of the last two syllables of a word after syncope.

3.2 Proto-Coptic vs Pre-Coptic Egyptian

Most of the works that have dealt with vocalization in Egyptian in the past have done so with the implicit or explicit assumption that they were using the Coptic evidence to reconstruct the vocalization of pre-Coptic Egyptian as *attested* in earlier Old/Middle/Late Egyptian hieroglyphic sources. However, as already occasionally observed by previous scholars (e.g. Edgerton 1947: 3–4; Greenberg 1969; Hintze 1980; Loprieno 1995: 61), this way of formulating the problem is somewhat imprecise. Rather, the variety that can be reconstructed on the basis of Coptic is more appropriately characterized as *Proto-Coptic*.

It is important to emphasize that Proto-Coptic should not be equated with the attested stages of pre-Coptic Egyptian. Like all proto-languages, Proto-Coptic is an abstract construct, designed to capture only those features of the earlier language that are recoverable from its descendants, in this case the Coptic dialects. As such, it represents a selective reconstruction rather than a complete linguistic system, and it is necessarily limited to what linguistic elements survived into the Coptic period.

Moreover, whatever historical variety of pre-Coptic Egyptian may have lain closest to Proto-Coptic is unlikely to have corresponded exactly to any of the lects attested in writing, but rather to an unrecorded spoken variety. This is because the written stages of Egyptian reflect highly conventionalized and diachronically conservative norms, often maintained across centuries and only partially responsive to ongoing phonological change. By contrast, Proto-Coptic is a retrospective abstraction that captures the common structural properties inherited by Coptic dialects, which necessarily derive from earlier spoken forms of the language. The phonological developments visible in Coptic therefore presuppose intermediate vernacular varieties that it is safe to assume were never fully fixed in writing, and which may have diverged from the contemporary written standards. Comparable situations are well known from other language families. For instance, Proto-Romance, i.e. the proto-language that can be reconstructed on the basis of modern Romance languages, differs in important respects from Classical Latin as recorded in writing. Similarly, the proto-language that can be reconstructed from modern spoken forms of Arabic cannot be straightforwardly identified with Classical Arabic, as the two differ substantially in multiple respects. At the same time, although Proto-Coptic is treated here as a distinct construct, it is nonetheless historically related to pre-Coptic Egyptian as attested in the written record. The reconstructed forms proposed for Proto-Coptic can therefore be expected to stand in a meaningful relationship to forms documented in earlier Egyptian. In many cases, however, the precise nature of this relationship still requires clarification.

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Addressing these issues clearly lies beyond the scope of the present paper and must be left to future research. For the purposes of this study, it is sufficient to emphasize that the reconstructions proposed here are Proto-Coptic in nature, rather than intended to reflect any specific attested stage of pre-Coptic Egyptian. That said, historically stable consonants reconstructed for Proto-Coptic are generally transcribed as they appear in pre-Coptic Egyptian forms. This choice is motivated both by considerations of clarity, as it facilitates the identification of cognate forms across stages, and by the fact that it allows a direct assessment of whether such etymological consonants played a role in the phonological phenomena discussed in this paper.

Proto-Coptic forms are indicated by a single asterisk *.

3.3 Pre-Proto-Coptic

I will occasionally refer to a Pre-Proto-Coptic stage of the language. This term should be understood as a hypothetical stage whose features precede those reconstructible for the Proto-Coptic to Coptic transition, but which can be postulated to account for patterns that are reconstructible within Proto-Coptic itself or inferred on the basis of extra-Coptic or extra-Egyptian evidence, such as comparison with other Afroasiatic languages. Pre-Proto-Coptic forms are indicated by a double asterisk **. In Pre-Proto-Coptic, the stress could fall on the final, penultimate, or antepenultimate syllable of the word before syncope.

Two features, in particular, are postulated in this article for Pre-Proto-Coptic:

1) Phonemic length in vowels: As discussed in Kilani 2025 (in this journal), I follow Vycichl in reconstructing phonemically long vowels at the Pre-Proto-Coptic level. Vocalic length at this stage would have played a role in determining stress placement in Proto-Coptic (and thus in Coptic), but the phonemic distinction in vowel length was likely lost by the Proto-Coptic stage, as it seems to play no role there. This earlier opposition is therefore unrelated to any vowel length that may have developed secondarily between Proto-Coptic and Coptic (see below on Coptic vowels written twice). While the presence of phonemically long vowels in Pre-Proto-Coptic can be reasonably argued for in certain contexts, other cases remain more ambiguous. In my opinion, the strongest evidence is provided by disyllabic words with stress on the second syllable, since, if the second syllable contained a short vowel, stress would be expected to retract to the first syllable. This can be summarized as follows:

i) **CV́CVC* < ***CV́CVC*
ii) **CVCV́C* < ***CVCV́:C*

Additional, more ambiguous cases are considered below.

2) Long/Geminate consonants[2]: As suggested by several scholars (e.g. Gardiner 1957: 207; Vycichl 1984: x and passim, 1963; Peust 1999: 160), it seems reasonable to assume that Pre-Proto-Coptic may have had geminate consonants, even though their presence

2 Note that the term 'geminate' is used here in its standard linguistic sense, to refer to a lengthened consonant. This should be distinguished from *reduplicated* consonants found in some Coptic and

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is often difficult to demonstrate directly, not least because they lost the gemination and merged with ungeminate consonants by the time of Coptic. In my view, the strongest evidence for their existence comes from irregularities in stress placement in Coptic and Proto-Coptic. In particular, I would argue that the presence of a geminate consonant can be reasonably assumed in cases where stress shifts between grammatical variants of the same word in ways that cannot be accounted for by the standard syllabic theory alone. For example, **CV́CCVt* represents the regular and expected Proto-Coptic pattern for the feminine counterpart of masculine **CV́CVC* nouns. However, a small number of masculine nouns of the type **CV́CVC* display feminine counterparts with the unexpected pattern **CVCV́CVt*. In my view, this atypical behaviour can be reconciled with the standard syllabic theory only by assuming that the second consonant was geminate in Pre-Proto-Coptic, thereby blocking the syncope of the second vowel when the feminine ending was added. Under this assumption, the development would be as follows:

i) m. **CV́CVC* < ***CV́CVC* — f. **CV́CCVt* < ***CV́CVCVt*
ii) m. **CV́CVC* < ***CV́C:VC* — f. **CVCV́CVt* < ***CVC:V́CVt*

Instances of trisyllabic words with stress on the second syllable, for which no grammatical variants are attested that could reveal alternative stress patterns, remain difficult to interpret, as they may reflect either a geminate consonant or a long vowel. In particular, an isolated form of the type **CVCV́CV(C/t)* could be interpreted either as:

**CVCV́CV(C/t) < **CVC:V́CV(C/t)*

or as:

**CVCV́CV(C/t) < **CVCV́:CV(C/t)*

In order to represent this ambiguity within a single Pre-Proto-Coptic form, I will indicate uncertain length by placing a *ˀ* after the length mark *ː* , i.e.:

***CVC:ˀV́:ˀCV(C/t) = **CVC:V́CV(C/t)* or ***CVCV́:CV(C/t)*

3.4 Root classes

The present article follows the model proposed in Kilani 2025 with respect to the definition of verbal root classes. Accordingly, rather than adopting traditional categories such as 3-literal, third-weak, second-geminate, and similar classifications, I employ the following definitions:

- *3-EtR*[3] : verbal roots that can be reconstructed with three strong consonants or with an initial weak consonant and two additional strong consonants
- *2*[+]*-EtR* : verbal roots that can be reconstructed with two strong consonants and likely a non-initial weak consonant, whose location is uncertain

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pre-Coptic Egyptian nouns and verbal forms, which are occasionally, and somewhat misleadingly, referred to as 'geminate' in the Egyptological literature.

3 *EtR* : 'Etymological Root'.

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Although in most contexts the non-initial weak consonant associated with *2⁺-EtR* roots does not survive in Coptic, in a few cases direct or indirect traces of it can be detected (see, for example, forms such as [A] ⲡⲁϫⲣⲉ |páwrə| – Kilani 2025: 152). As a result, it is reasonable to reconstruct this consonant at the Proto-Coptic level. In the rare cases where the exact position of this consonant can or needs to be precisely indicated, *X* will be used. In most cases, however, the exact position cannot be established; it can only be assumed to have occurred either before or after the second strong consonant of the root. In these cases, I will transcribe it with a cross ˣ placed over the second consonant, i.e.: *CVC̽V* = *CVCXV* or *CVXCV*.

3.5 Conventions in Transcription

In the transliteration of Proto-Coptic and pre-Coptic Egyptian:

I use an Egyptological transliteration based on Gardiner's, with *i̯* as a conventional sign for the consonant corresponding to 𓇋 and *q* for 𓎛.

In the transcription of Coptic:

- Dialects are indicated within square brackets [ ] ; the abbreviations used are the same as those employed by Crum in his dictionary (Crum 1939).
- I use the notation | | to signal quasi-phonological *interpretations* of Coptic forms. I describe these as "interpretations" rather than "transcriptions" or "reconstructions" for the following reasons: they are not reconstructions, which would be marked with *, since they correspond to attested forms rather than hypothetical ones; they are also not fully phonological transcriptions, which would be marked with / / or [ ], because the phonology of historical Coptic, while generally understood, is less securely established than that of a living language and continues to be debated in certain respects. The forms presented in | | therefore reflect *my interpretations* of the likely pronunciation of the corresponding Coptic forms (based mainly on Loprieno 1995; Peust 1999). It should be emphasized, however, that these *interpretations* are not definitive, and other scholars may disagree with particular details (on these issues, see Müller 2011).
- Besides the obvious equivalences, the following conventions need mention: ⲱ = |o|, ⲟ = |ɔ|, ⲏ = |e|, ⲉ in stressed position = |ɛ|, ⲉ in unstressed position = |ə|, ⲓ as vowel in stressed position = |i|, ⲓ as vowel in unstressed position = |i̯|,[4] ⲓ as consonant = |j|, ⲟⲩ as vowel = |u|, ⲟⲩ as consonant = |w|, ⲕ = |kʲ|, ⲭ = |č|, ϩ = |h|, ϧ = |h₂|
- Stress is marked with ´
- Vocalic/Syllabic consonants are marked with ̩

4 The use of i here is partially conventional, and it is meant to represent an unstressed and likely reduced (mostly final) vowel in those words and dialects where it is spelled with ⲓ rather than the more common ⲉ.

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– Vowels written twice are interpreted as long, and indicated with : e.g. ⲁⲁ = |aː|. Note that other scholars interpret double vowels as indicating a glottal stop, e.g. ⲁⲁ = |aʔ|. The use of | | signals explicitly that these are interpretations subject to debate, not established phonological facts.

– The sign ᵊ indicates the potential presence of an unstressed vowel in a position where such vowel is not explicitly indicated in the corresponding Coptic spelling.

In reconstructions:

– *C* = unspecified/any consonant

– *V* = unspecified/any vowel

– $\mathcal{X}$ and ^{x} = hypothetical, undetermined weak consonant in *2⁺-EtR* roots

– *ʔ* = a weak consonant derived from the weakening of an earlier, attested *r*, *ʿ*, *ɜ*.

– *ẏ* = unspecified semiconsonant/glide = *j*, *w*, *ʔ*

– I mark with a dot below the letter ạ by analogy with the convention used to indicate uncertain readings in manuscript transcriptions, those phonemes (mostly vowels) that cannot be reconstructed directly from the internal evidence alone, but which can be postulated with relative confidence on the basis of analogy or extra-Egyptian-Coptic evidence (e.g. comparison with other Afroasiatic languages).

4 Proto-Coptic Vowels: Deconstructing the current model

Current models reconstruct three phonemic vowels at the pre-/Proto-Coptic level, namely **a*, **i*, **u*. There are two main arguments that have been used to support the reconstruction of a three-vowel system for pre-Coptic Egyptian, and these arguments implicitly, and only rarely explicitly, underlie most current approaches to the issue. The first is the observation that Proto-Semitic is commonly reconstructed with three vowels. The second is the apparent occurrence of three vowels in second-millennium cuneiform transcriptions of Egyptian words. Both of these arguments are methodologically problematic.

4.1 The Semitic bias

Semitic is the best known and by far the most studied branch of the Afroasiatic family. Moreover, while Egyptian is attested earlier than any Semitic language, Semitic languages such as Akkadian and Eblaite provide the earliest attestations of vocalization in the entire Afroasiatic family. As a result, Semitic languages and reconstructed Proto-Semitic played a very prominent role during the heyday of Afroasiatic linguistics. In Egyptology, this Semitic bias went even further, as several early twentieth-century Egyptologists believed that Egyptian was itself a Semitic language, albeit one that had diverged from the rest of the family at a very early stage.

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Significantly, this was the opinion of Sethe (1923: 175), the first scholar to propose pre-Coptic vowels, as stated in the very article where he proposed his model.[5] Within this framework, reconstructing the phonology of pre-Coptic Egyptian could be seen as tracing how the supposedly original Proto-Semitic phonemes developed in Egyptian. This approach is evident in Albright's work (1923), the second pioneer in studying Egyptian vocalization.

Now, Proto-Semitic has long been assumed, already in Sethe's time, to have three phonemic vowels that could appear both in a long and in a short form, **a/ā*, **i/ī*, and **u/ū*. It was therefore natural for scholars of the period to assume that the pre-Coptic Egyptian vocalization ultimately matched these three vowels. In fact, this assumption was so self-evident that it was never argued for in their works and appeared simply as an implicit prerequisite.

Later scholarship, however, has increasingly questioned the idea that Egyptian derives from Semitic, and the consensus has gradually shifted away from this view (see Almansa-Villatoro and Štubňová Nigrelli 2023 for a recent overview). Even those who still argue for a close relationship now generally treat Egyptian and Semitic as sister branches within the wider Afroasiatic family, rather than as a direct line of descent. Moreover, research in recent decades on the African branches of the Afroasiatic family has shown that the Semitic vocalic model based on a three-way opposition is far from ubiquitous, typical, or self-evident across the family.[6] Specifically, recent scholarship has advanced a range of reconstructions of Proto-Berber favouring a mixed system characterized by three or four long vowels and a reduced set of only two short vowels (Kossmann 2020: 34–37). For Proto-Chadic, various configurations have been proposed, ranging from maximalist models with four phonemic vowels without length contrast to minimalist ones featuring only a binary opposition either between two vowels or between the presence and absence of a single vowel (Wolff 2022: 2025). As for Cushitic, most branches are generally reconstructed with five vowels without length distinction (Mous 2012: 353), although alternative proposals have been made (Bender and Hudson 2020: 133, 165). The Agaw languages have been occasionally reconstructed with six or seven vowels (Ehret 1987: 10), although more recent scholarship tends to favour a five-vowel model (Appleyard 2006: 11). The situation in Omotic is broadly similar to that of Cushitic, and this resemblance has been suggested to reflect areal contact (Theil 2012).

This diversity is also evident in Proto-Afroasiatic reconstructions, where competing models propose vowel systems of differing sizes, ranging from a minimum of two vowels to substantially larger inventories (see e.g. Diakonoff et al. 1987; Ehret 1995; Orel and Stolbova 1995; Levin 2003). Yet none of this appears to have reached Egyptology, where the three-vowel model required by the early twentieth-century view of Egyptian as a Semitic language still persists.

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5 See page 175: "…auf den Urzustand des ägyptischen Vokalismus gekommen sind, auf den Zustand, wie er bei der Abzweigung des Ägyptischen von dem semitischen Sprachstamm herrschte,…"

6 In fact, the three-vowel system has recently been questioned also for Proto-Semitic. In particular, Ahmad Al-Jallad argued at a workshop in Basel in May 2026 (https://mkilani.github.io/lcde-workshop/), that the Semitic verbal system can be analysed as reflecting only two functional phonemic vowels.

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4.2 Cuneiform transcriptions

While the presumed Semitic origins of Egyptian have been the main argument for assuming a three-vowel model, it has long been claimed or assumed that Late Bronze Age cuneiform transcriptions of Egyptian words offer clear evidence for their existence. This, however, is not the case.

The borrowing of words into a foreign language, or their transcription in a foreign script, is rarely a trivial process. Rather, it is usually shaped and constrained by different factors, and this applies also to cuneiform transcriptions of Egyptian words. Two issues need to be considered.

The first issue is that the ability of any script to represent foreign sounds is constrained by its own graphemic inventory. This is not a problem when the foreign sounds that need to be transcribed are identical or sufficiently close to the sounds underlying the script itself. The issue arises, however, when a sound to be transcribed lacks obvious correspondences in the receiving script. In such cases, users of the script will employ one or more of its existing graphemes to approximate the foreign sound. For instance, Akkadian cuneiform had no dedicated sign to render the sound [ə], and graphemes corresponding to other vowels had to be used instead when transcribing foreign names containing it. This is clear, for example, from Akkadian texts from Ugarit, where the spellings *niq$_2$-**me**-pa-a*, *niq$_2$-**mu**-pa*, and *niq$_2$-**ma**-pa-a* are attested, all attempts to approximate a name that was likely pronounced *niqməpa* at the time (Howard 2023: 50).

In general, recent scholarship on Akkadian transcriptions of foreign words has clearly shown their idiosyncratic and unpredictably approximative nature. This is evident, for instance, when we compare Akkadian transcriptions of foreign names with their known forms in the source languages (Waerzeggers and Gross 2024). All this considered, it would be quite surprising if cuneiform Akkadian were truly providing accurate, one-to-one transcriptions of Egyptian vowels. Or, at the very least, it would be extremely optimistic to assume this as a default.

The second issue affecting cuneiform transcriptions concerns the very nature of what is being transcribed. When it comes to the phonology of languages, modern linguistics distinguishes between a *phonemic* and a *phonetic* reality. The phonemic reality consists of the abstract sound units that speakers of a language perceive as contrastive and meaningful, whereas the phonetic reality concerns the concrete acoustic realizations of those units in actual speech. Now, scholars working on contact linguistics generally agree that the primary interface in linguistic contact, at least in oral contexts, is the phonetic reality rather than the underlying phonemic one, and as a result, borrowings are mediated primarily by the listener's own *phonetic* system and only secondarily by their *phonemic* categories (Uffmann 2015: 644, 645–47, 652–57; see also Kilani 2023). In concrete terms, this means that phonemic contrasts that are distinctive in the source language can be neutralized and merged when words are borrowed, if they are not phonemically distinct in the receiving language. Conversely, phonological differences that are not phonemic in the source language but exist at the phonetic level may be adopted or transcribed as distinct sounds when they align with true phonemic oppositions in the receiving language. Numerous cross-lin-

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guistic examples of both phenomena are attested. For instance, the wide range of vowels of some European languages, such as French or English, is systematically reduced to a three-way opposition corresponding to the three phonemic vowels of Arabic when words from these languages are borrowed or transcribed into Arabic.

Greek transcriptions of Egyptian words provide another instructive example: we know that Coptic *u* and *o* (e.g. [S] ⲱ |o|, ⲟⲩ |u|) were contextual allophones of a single underlying phonemic vowel, usually assumed to derive from an earlier **a*, namely **a* > *u* after nasal and **a* > *o* in other contexts.

Yet, Greek transcriptions (and indeed Greek-derived Coptic orthography as well) represent these two phones in several different ways, sometimes using the same vocalic graphemes for both and other times distinguishing them with different vocalic graphemes:

- Greek ω ~ Coptic ⲱ |o| < Egyptian **a*
  E.g.: Gr. ωρος : 'Horus' ~ Coptic [S] ϩⲱⲣ |hór(ᵊ)| < **ḥarV* (Vycichl 1984: 307)
- Greek υ ~ Coptic ⲱ |o| < Egyptian **a*
  E.g.: Gr. παυνι : '10th month' ~ Coptic [S] ⲡⲁⲱⲛⲉ |paónə| < **pV-ỉánVt* (Peust 1999: 311)
- Greek υ ~ Coptic ⲟⲩ |u| < Egyptian **a*
  E.g.: Gr. (σεβεν)νυτος : 'god' in the name of the city of Samannoud ~ Coptic [S] ⲛⲟⲩⲧⲉ |nútə| < **náṯVr* (Vycichl 1984: 145)
- Greek ου ~ Coptic ⲟⲩ |u| < Egyptian **a*
  E.g.: Gr. ανουβις[7] : 'Anubis' ~ Coptic [S] ⲁⲛⲟⲩⲡ |anúp(ᵊ)| < **ỉVnábV* (Vycichl 1984: 13)
- Greek ω ~ Coptic ⲟⲩ |u| < Egyptian **a*
  E.g.: Gr. αμμων (et sim.)[8]: 'Amoun' ~ Coptic [S] ⲁⲙⲟⲩⲛ |amún(ᵊ)| < **ỉVmánV* (Vycichl 1984: 10)

If we were to reconstruct Egyptian vowels on the basis of Greek transcriptions alone, we would likely posit three distinct phonemic vowels corresponding to Greek ω, υ, and ου. If, instead, we reconstructed them on the basis of each different Coptic–Greek correspondence, we might even propose as many as five different vowels. Yet all current models realistically reconstruct only a single phonemic vowel **a* in pre-Coptic Egyptian, which later developed into two allophones in Coptic but was variously adapted or transcribed as ω, υ, and ου in Greek.

The same may well have been true for cuneiform transcriptions of Egyptian words. Egyptian could have had only two phonemic vowels with three or more phonetic realizations, and it may have been these phonetic realizations, rather than the underlying phonemic vowels, that were represented in cuneiform. Nothing in the cuneiform evidence itself excludes such a possibility.

It is also important to stress that the consistent spelling of a particular vowel in cuneiform does not necessarily imply an accurate representation of the corresponding Egyptian

7 Also αννυφις (Vycichl 1984: 13).

8 Also αμουν in Herodotus (Vycichl 1984: 10).

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vowel at the phonemic level. An illustrative parallel is provided by the name of the Lebanese capital, Beirut, which is consistently spelled with <e> in most, if not all, European languages. This, however, does not mean that (Lebanese) Arabic has a phonemic vowel /e/. On the contrary, the phonemic form of the city's name is /baj.ruːt/, with /a/. The widespread use of spellings with <e> in European languages can instead be explained as reflecting the phonetic realization of the diphthong /aj/ as [ej] in spoken Lebanese Arabic, the likely source of the borrowing, possibly reinforced by an orthographic standardization[9] of the name in official contexts across Europe. In other words, the consistent use of <e> in European spellings is not sufficient to claim that Lebanese Arabic has a phonemic /e/. It merely suggests that some aspect of the phonetic realization of this particular word and/or of European orthographic practices, led to the conventionalization of the spelling <e> in European languages. Similarly, the fact that, for instance, the Egyptian word *mꜣꜥt*, which frequently appears in Egyptian royal names, is predominantly spelled with <u> in cuneiform transcriptions cannot automatically be taken to imply that Egyptian had a phonemic /u/, as this is only one of several possible explanations. Rather, it simply suggests that the phonetic realization of this particular word, and/or the way cuneiform spellings of Egyptian royal names were learned and transmitted among cuneiform scribes, led to the conventionalization of a cuneiform spelling with <u> for this specific lexical item.

It is thus impossible to say whether this transcription practice reflects a genuine phonemic /u/ or not, because the cuneiform evidence alone is not enough to exclude the possibility that this is instead a context-dependent phonetic realization that cuneiform scribes simply perceived as sounding close to their <u>. For instance, in the specific case of *mꜣꜥt*, it is well known that in Egyptian-Coptic /m/ and /n/ cause a following [o] to be realized as [u], suggesting a labialization process triggered by nasal consonants that may also have affected other vowels in similar contexts. Alternatively, the cluster *ꜣꜥ* may have conditioned the phonetic realization of the preceding vowel making it sound like a back vowel compatible with cuneiform <u>. One can think of languages such as Irish, where vowels show strong coarticulatory conditioning by nearby palatalized versus velarized consonants, with the two high and mid short vowels surfacing as [i] and [e] respectively in palatalized environments, and as [u] and [o] in velarized ones (Ó Siadhail 1989: 33–36, 83–89).

5 A data-driven, inner-Egyptian-Coptic approach to pre-Coptic vowels

If the presence of three phonemic vowels in Semitic is not sufficient to assume the same for Egyptian, and if cuneiform transcriptions alone are not adequate evidence either, then how can we determine how many phonemic vowels pre-Coptic had?

Methodologically, the best approach is to rely only on internal Coptic evidence, and there are two ways to do so. The first is to determine the minimum number of vowels that must be reconstructed for Proto-Coptic to yield coherent results. The second is to identify minimal pairs that imply distinct vowels at the lexical or morphological levels.

9 One may also wonder about the role of Ottoman Turkish in this process of transmission.

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5.1 Reconstructing vowels in Proto-Coptic

The first approach to reconstructing the vowels of Proto-Coptic is to apply the comparative method, beginning with the Coptic evidence and examining distributional patterns that may reveal both underlying phonemic oppositions and regular allophonic realizations. To this end, it is important to note that Coptic vowels can be divided into two major classes depending on the type of syllable in which they occur: closed (CVC) or open (CV). A further distinction must be made between stressed and unstressed vowels, since the former clearly preserve different reflexes of distinct phonemic Proto-Coptic vowels, whereas the latter have generally merged and been reduced to a single, indistinct vowel. Consequently, only stressed vowels are considered here.

5.1.1 Stressed vowels in closed syllables

There is general agreement that only two distinct vowels can be reconstructed in closed syllables based on the Coptic evidence. Their primary realizations in the main Coptic dialects are summarized in Table 1.

Table 1 | Distribution of the two phonemic vowels and their allophones in closed syllables in the main dialects of Coptic (Osing 1976, 11).

		Sahidic	Bohairic	Akhmimic	Fayyumic
Vowel 1	Basic Form	ⲟ |ɔ|	ⲟ |ɔ|	ⲁ |a|	ⲁ |a|
	Main Allophones[10]	ⲁ |a|	ⲁ |a|	ⲟ |ɔ|	ⲉ |ɛ|
Vowel 2	Basic Form	ⲁ |a|	ⲁ |a|	ⲉ |ɛ|	ⲉ |ɛ|
	Main Allophones[11]	ⲉ |ɛ|	ⲉ |ɛ|		

Vowel 1 is usually reconstructed as **a*, with some dialects showing the cross-linguistically common development **a* > ⲟ |ɔ|. Vowel 2 lacks an established transcription: Osing (1976: 11, and passim) uses **e* whenever he cannot distinguish between his supposed **i* and **u*, but I prefer **E*: the capital letter indicates that this is not a claim that the vowel was /e/, but a conventional transcription of a vowel contrasting with **a*, perhaps realized as a central, high, or front vowel.

While there is general agreement that only these two vowels can be reconstructed in closed syllables,[12] the widespread assumption of a Semitic-like three-vowel system has led previous scholars (e.g. Osing 1976: 11–18; Loprieno 1995: 38–39; Allen 2020: 46) to postulate that Proto-Coptic Vowel 2 resulted from the merger of the two presumed Pre-Proto-Coptic vowels **i* and **u*.

10 Primarily before ʿ, ḥ, ḫ, ẖ, and occasionally h (see Osing 1976: 11).

11 Multiple conditioning factors; for a discussion, see Osing 1976: 11–18.

12 Although some attempts to distinguish two proto-vowels **i* and **u* at least in some contexts have been occasionally suggested (see e.g. Osing 1976: 11, and passim), none has led to any clear or coherent rule or set of correspondences that could be applied across multiple words.

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In a second step, cuneiform sources have then been used to seek earlier reflexes of these vowels in the ancestors of words that show Vowel 2 in Coptic. The Coptic data, however, provide no evidence for more than a single Vowel 2, and, in light of the discussion above, the cuneiform material is insufficient to overturn this conclusion. As argued earlier, the use of both <i> and <u> signs in cuneiform to render the supposed Vowel 2 does not prove that this vowel derives from two distinct pre-Proto-Coptic vowels **i* and **u*. Rather, it may simply indicate that Vowel 2 had no exact equivalent in Akkadian, but was realized in a way that lay somewhere between the values represented by cuneiform <i> and <u> (e.g. something similar to [ə], [ɨ], [ʉ], or perhaps even [y]), or that the phonemically single Vowel 2 had two (or more) allophonic realizations in Egyptian itself that cuneiform scribes perceived as close to the Akkadian vowels written with <i> and <u> in cuneiform.

5.1.2 Stressed vowels in open syllables – the mainstream model

The situation in Coptic open syllables is similar although more complex. In Sahidic, four vowels can appear in open syllables, namely [S] ⲱ |o|, ⲟⲩ |u|, ⲓ |i|, ⲏ |e|. Other dialects have similar profiles. The vowel [S] ⲟⲩ |u| is generally considered as a mere allophone of [S] ⲱ |o| primarily after nasals; both are derived from a Proto-Coptic vowel **a* (see e.g. Osing 1976, 11). The case of [S] ⲓ |i| and [S] ⲏ |e| is more complicated. The assumption that pre-Coptic Egyptian had two vowels **i* and **u* has led scholars to identify their reflexes in [S] ⲓ |i| and [S] ⲏ |e|, usually proposing **i* > [S] ⲓ |i| and **u* > [S] ⲏ |e| (Osing 1976: 19; Loprieno 1995: 38–39; Allen 2020: 46). However, the reality of the data contradicts in many ways these tempting correspondences.

First, if these represented genuinely distinct phonemic vowels, [S] ⲓ |i| and [S] ⲏ |e| would not be expected to emerge from the same underlying vowel in different morphological contexts. Nevertheless, this is exactly what is observed in pairs such as [L] ϩⲉϩ |mˢhé| < **mVḫ́É:ʔẏạt* : 'northwind' and in a form (nisba) derived from it [S] ϩⲉⲓⲧ |mˢhít(ˢ)| < **mVḫ́É:ʔẏtEj* : 'north'; in these two forms the vowel was exactly the same, but in the first it emerges as ⲏ |e|, while in the second as ⲓ |i|.

Moreover, if the traditional analysis were correct, one would expect cuneiform transcriptions, or Coptic words derived from Semitic loanwords to show unambiguous **u* > [S] ⲏ |e| and **i* > [S] ⲓ |i| derivations. This is not the case, and instead all the following developments appear to be attested:

- **u* > [S] ⲏ |e| – [S] ϭⲓⲛⲏⲣⲉ |kʲinérə| : 'string instrument' supposedly from **kinurV,* a Semitic loanword; see e.g. Heb. *kinnôr* (Vycichl 1984: 342; Hoch 1994: 324).
- **u* > [S] ⲓ |i| – e.g. Cpt. [S] ⲧⲣⲓⲣ |tˢrír| : 'oven' supposedly from **tVrurV*, a Semitic loanword (Vycichl 1984: 221; Hoch 1994: 359).
- **i* > [S] ⲓ |i| – e.g. Cpt. [S] ϥⲓⲧ |pˢsít(ˢ)| : 'nine' supposedly from **pVsiḏiẏ* transcribed as *pi/e-ši-iṭ* in a cuneiform tablet from Amarna (Vycichl 1984: 248).
- **i* > [S] ⲏ |e| – e.g. Cpt. [S] ⲣⲏ |re| : 'sun' supposedly from **riʿV*, transcribed as *-ri-a* (et sim) in cuneiform (Vycichl 1984: 170).

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Scholars have often tried to explain these variations as being the result of allophonic contextual developments. However, these explanations often appear *ad hoc*, or fail to explain the totality of the data.

For instance, Osing (1976: 460–61) has argued that one of the conditions for **i* > ⲏ |e| is the presence of the consonant ʿ after the vowel. This supposed allophonic development, however, is based on only a single word,[13] namely Coptic [S] ⲣⲏ |re| : 'sun', which appears as *-ri-a* (et sim) in contemporary cuneiform transcriptions of names of New Kingdom kings. Moreover, counterexamples were known to Osing and led him to state that this proposed "rule" is actually not an obligatory one;[14] i.e. we can say it is not a rule at all (Osing 1976: 461). Other scholars have taken a more cautious approach, accepting these forms as allophonic variations but refraining from proposing any rule to account for them. Peust (1999: 224), for instance, notes that **i* > [S] ⲏ |e| occurs under "special conditions", but does not attempt to specify what those conditions are.

Similarly, scholars have suggested that **u* > [S] ⲓ |i| is an allophonic realization that occurs before *r* (Osing 1976: 19). This rule could explain cases like Coptic [S] ⲧⲣⲓⲣ |tˁrír| : 'oven' < **tVrurV* (Vycichl 1984: 221; Hoch 1994: 359), but it is easily contradicted by, e.g. CⲏC statives, where the vowel is ⲏ |e| even before *r* ; see e.g. the stative of verbs like [S] ⲥⲏⲣ+ |sér| < √*sr̓*: '(to) spread'.

In fact, if we exclude Coptic words derived from loanwords, it appears that the presence of ⲏ |e| or ⲓ |i| does not correlate with any consonant, but in several cases strongly correlates with, and can often be predicted from, the *morphological* context in which the vowel appears. This observation will be further explored below.

5.1.3 Stressed vowels in open syllables – a less successful model

Before moving forward, it is worth noting here that a few scholars who support the reconstruction of phonemically long vowels at the pre-Coptic stage have proposed alternative models in which vowel length influenced the realization of Coptic vowels. One early proponent of this approach was Lambdin (1958: 191–92), who suggested that **u* / **ū* > ⲏ |e| and **ī* > ⲓ |i|, whereas **i* > ⲏ |e|. However, his proposal was never elaborated in detail and received little attention within the field.

The second proposal was put forward by Vycichl (1957: 216–18, 1959: 51–54; see also Peust 1999: 231), who suggested that **i* > ⲓ |i|, **u* > ⲏ |e|, but **ī* > ⲏ |e|, **ū* > ⲓ |i|. Vycichl introduced this rather peculiar model in some of his earlier works, but it seems that he changed his mind and moved away from it in later years (see Peust 1999: 231).

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13 The other example cited by Osing, namely *wʿw* : 'soldier' which is transcribed as *ú-i-ú* in early cuneiform transcriptions and as *ú-e-ú* (et sim.) in later ones (Loprieno 1995: 39), is not valid, as the word does not survive in Coptic and could reflect other phenomena, such as different transcription traditions within cuneiform or different Egyptian dialects with varying phonetic realizations of that vowel as sources of the two cuneiform forms.

14 "…nicht obligatorisch war" (Osing 1976: 461).

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5.2 Morphological, lexical, and pattern minimal pairs

As noted, a traditional comparative-method approach allows for the reconstruction of two proto-vowels in CVC syllables, but it fails to provide a coherent explanation for the presence and distribution of the vowels ⲓ |i| and ⲏ |e| in open syllables. In this case, an alternative strategy to approach the problem is to try to determine the number of underlying phonemic vowels. To do this, one can try identifying minimal pairs at the lexical or morphological level.

5.2.1 Lexical minimal pairs

Lexical minimal pairs are pairs of lexical items that differ by only one phoneme in an otherwise identical phonological environment. For example, the Italian vowels /o/ and /ɔ/ can be shown to be phonemically distinct because they are the sole distinguishing feature between the words <botte> /bɔtːe/ : 'barrel' and <botte> /bɔtːe/ : 'blows'.

The case of Coptic is somewhat more complex because the question at stake is not whether ⲏ |e| and ⲓ |i| had full phonemic status in Coptic itself, but rather whether they derive from two distinct proto-vowels with phonemic status in Proto-Coptic. Demonstrating this would require identifying a minimal pair consisting of two lexical items that have identical forms in Coptic except for the stressed vowel, one containing ⲏ |e| and the other ⲓ |i|, *and* that also share the same consonants and the same unstressed vowels at the pre-Coptic stage.

In practice, suitable cases are extremely difficult, if not impossible, to identify. Pre-Coptic weak consonants *j*, *w*, ʿ, ʾ and geminate consonants were frequently lost or simplified in ways that left no distinctive trace in Coptic, making their reconstruction at the Proto-Coptic level highly uncertain. Restricting the analysis to words with three strong consonants, which can therefore be assumed not to contain hidden weak consonants, does not resolve the problem. On the one hand, it cannot be excluded that those appearing as single consonants in Coptic or pre-Coptic were in fact distinct phonemes characterized by secondary articulations that may have affected vowel realization. For instance, Peust (1999: 107) has convincingly argued that at least some of the pre-Coptic velar plosives usually written with the hieroglyphs 𓎛, 𓎡, 𓎝 must have had hidden phonemically distinct consonants that were characterized by secondary features not represented in writing (e.g. aspiration, labialization).

On the other hand, as said, it is likely that at some point in its history, pre-Coptic Egyptian had geminate consonants (as is the case for many other Afroasiatic languages), which however became indistinguishable from simple ones in Coptic. While this process may have left no distinctive trace on the consonants themselves, there is no way to rule out differential effects on adjacent vowels.

Similarly, it has been proposed (e.g. Vycichl 1984: x; see also Kilani 2025: 141) that early stages of pre-Coptic Egyptian possessed phonemic vowel length, comparable to that found in Semitic languages. Vowel length would originally have played a role in determining stress placement, but was later entirely neutralized, resulting in an allophonic differentiation of the vowels conditioned by whether the syllable was open or closed. If

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this scenario is correct, then in principle the contrast ⲓ |i| versus ⲏ |e| could reflect an earlier opposition in vowel length affecting a single phonemic vowel, /E/ versus /Ē/ (or more, as some scholars have suggested − see above).

Finally, true minimal pairs would also need to share the same unstressed vowels, since vowels in neighbouring unstressed syllables could in principle have influenced the realization of the stressed vowel through processes such as vowel harmony or metaphony.[15] However, as said, unstressed vowels are often very difficult to reconstruct, especially in nominal forms where no paradigmatic analogy can be applied (see below).

All these issues can be illustrated with a practical example: the Coptic words [S] ⲥⲓϥⲉ |sífə| 'tar (tree), cedar' and [S] ⲥⲛϥⲉ |séfə| 'sword, knife' may at first sight appear to form a perfect minimal pair, as in Coptic they differ only in the quality of their stressed vowel. In fact, as in the case of Italian /botːe/ vs /bɔtːe/, these forms could be used to argue that in (written) Coptic, ⲓ and ⲏ indeed worked as different phonemes, as they are the only formal element distinguishing these two words. However, when their pre-Coptic forms are examined, their Old Egyptian consonantal bases prove to be different: *sf̲t* for the former and *zf̲t* or *zf.t* for the latter (Vycichl 1984: 204). A first alternative explanation could thus be that these words originally contained a single phonemic vowel **E* in the first syllable, and that the different initial consonants *s*- and *z*- conditioned two allophonic realizations, which later emerged as ⲓ |i| and ⲏ |e| in Coptic. This, however, is not the only possibility.

Even setting aside the difference between *s*- and *z*-, the pre-Coptic ancestors of [S] ⲥⲓϥⲉ |sífə| and [S] ⲥⲛϥⲉ |séfə| may have differed in at least three additional respects. First, both words were likely already characterized in Proto-Coptic by a two-syllable structure with an unstressed vowel in the second syllable, i.e. **CV́CVC*. However, it is impossible to determine whether this vowel was identical in the two words, and any variation may have influenced the development of the stressed phonemic vowel in the first syllable. Moreover, if Pre-Proto-Coptic indeed had geminated consonants, the Pre-Proto-Coptic ancestors of both these words could be reconstructed in two different ways: ***CV́CVC*, ***CV́C:VC*. In addition, it has been suggested that Proto-Coptic **CV́CVC* nouns may, in some cases, derive from Pre-Proto-Coptic ***CV́CC* forms (Vycichl 1984: x); if so, there would be a third option to consider. Any of these potential differences, which are at present impossible to verify, may have created a conditioning environment that triggered distinct realizations of a single phonemic vowel.

This problem is not limited to Coptic nouns displaying the Cⲏ/ⲓC(ϩ)C / Cⲏ/ⲓCϩ < **CV́CVC* patterns, but also extends to the other two Coptic nominal patterns that show variants with both ⲏ |e| and ⲓ |i|, namely C(ϩ)Cⲏ/ⲓC(ϩ),[16] and Cⲏ/ⲓC(ϩ)[17] < *2*$^{+}$*-EtR* roots.

At first glance, the coexistence of these variants might suggest the presence of distinct underlying phonemic vowels. However, this interpretation appears unwarranted once all possible explanations are considered. In particular, the pattern C(ϩ)Cⲏ/ⲓC(ϩ) < Proto-

15 For a crosslinguistic parallel see for instance the case of Italian, where in many regions the phonemic vowels /e/ and /o/ are phonetically realized as [i] and [u], and /ɛ/ and /ɔ/ are realized as [e] and [o] when followed by unstressed [-i] or [-u] (Recasens 2021: 129–32).

16 E.g. [S] ϭⲣⲓⲛ |kᵖrím(ᵊ)| : 'bit', 'fragment' vs [S] ⲧⲃⲏⲗ |tᵖbél(ᵊ)| : 'fold (for sheep)'.

17 E.g. [S] ϩⲓⲛ |hín| : 'hin-vessel' vs [S] ⲥⲏⲧ |sét(ᵊ)| : 'tail'.

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Coptic **CVCV́CVy̆/t*, might in fact be derived from at least three distinct Pre-Proto-Coptic patterns, ***CVC:V́CVy̆/t* , ***CVCV́:CVy̆/t*, or in principle even ***CVC:V́:CVy̆/t*. Moreover, each of them contains two unknown unstressed vowels that could have influenced the realization of the stressed vowel in different ways.

The situation is even more complex for Cн/ıC(ϵ) forms derived from *2⁺-EtR* roots. Although these words have only two strong consonants in Coptic, they may once have included a third, weak consonant that was never recorded in writing, and whose precise position relative to the stressed vowel is impossible to determine (see Kilani 2025 on this). As a result, masculine Coptic Cн/ıC(ϵ) < *2⁺-EtR* could be interpreted as deriving from at least five different basic patterns in Proto-Coptic: **CV́XVC*, **CV́CVX*, **CVXV́CV(y̆)*, **CV́XCV(y̆)*, **CV́CXV(y̆)*. To these, we can add at least five more potential patterns in Pre-Proto-Coptic: ***CV́X:VC*, ***CV́C:VX*, ***CVX:V́CV(y̆)*, ***CVXV́:CV(y̆)*, or even ***CVX:V́:CV(y̆)*. Feminine Coptic Cн/ıC(ϵ) < *2⁺-EtR* could instead derive from at least three Proto-Coptic patterns, **CVXV́CVt*, **CV́XCVt*, **CV́CXVt*, with at least three more Pre-Proto-Coptic ones, ***CVX:V́CVt*, ***CVXV́:CVt*, or even ***CVX:V́:CVt*. And all of these involve one or two unknown unstressed vowels.

Taken together, this evidence shows that there is no pair of Coptic words that can be interpreted as an unambiguous lexical minimal pair, leaving open the possibility that the н/ı variation may simply reflect contextually determined allophonic variants of a single phonemic vowel.

5.2.2 Morphological minimal pairs

While lexical minimal pairs are clearly hopeless in Coptic, morphological minimal pairs have a better chance of being informative. The concept of morphological minimal pairs as indicator of phonemic vowels is clearly illustrated by Proto-Semitic, where the phonemic status of its three reconstructed vowels is demonstrated by their coexistence, in a three-way morphological opposition, namely case endings, as illustrated in Table 2 (Huehnergard 2019: 60).

Table 2 | Three-vowel phonemic opposition in Proto-Semitic case endings.

	Case	Example
Sg. Nom.	**-u*	**t'a:b-u-m*
Sg. Gen.	**-i*	**t'a:b-i-m*
Sg. Acc	**-a*	**t'a:b-a-m*

Similarly, morphological contrasts involving only a binary opposition can be observed in Chadic, and these have been used to support the reconstruction of a two-vowel system in Proto-Chadic (Wolff 2008: 47–49).

The same approach can be extended to Coptic. Yet, in order to do so, we first need to identify vowel oppositions that can be interpreted as morphological in nature, rather than merely deriving from non-morphological factors (such as, for example, lexically defined stem vowels in Semitic verbs). In particular, vocalic oppositions can be interpreted as reflecting morphological oppositions in two cases: first, when the same word displays two variants of the same morphological form differing only in one vowel; and second, when

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oppositions in vowels can be clearly correlated with oppositions at the morphosemantic level. Coptic provides three grammatical forms in which we can observe such vowel-encoded morphological oppositions in open syllables.

Significantly, in all three cases the morphological opposition appears to be binary, with one member of the pair containing a vowel derived from **a*, while the other contains ⲏ |e| (or ⲉ |ɛ| in closed syllable). By contrast, I have not identified any cases in which Coptic ⲏ /e/ and ⲓ /i/ form a morphologically meaningful contrast, nor any evidence supporting a three-way opposition among ⲏ |e|, ⲓ |i|, and **a*. This evidence therefore does not support a three-vowel system, but instead points to a binary opposition between only two proto-vowels, namely **a* and **E*. The principal cases are outlined below.

I. Plurals

A few Coptic nouns preserve morphologically distinct plural forms. While most of these nouns have only one plural, characterized either by a vowel derived from **a* or by the vowels ⲉ |ɛ| or ⲏ |e|, a small number preserve two plural forms displaying both vocalizations. For example:[18]

1	[S] ⲉϩⲟⲟⲩ |əhɔ́w(ᵊ)| < **i̯V̓ḥáy̯wV*	: ‘cows’ < ***i̯V̓ḥá:y̯wV*	2	[S] ⲉϩⲏⲩ |əhéw(ᵊ)| but also [S] ⲉϩⲉⲩ |əhéw(ᵊ)| < ***i̯V̓ḥÉ(y̯)wV* < ***i̯V̓ḥÉ:(y̯)wV*	: ‘cows’ (possibly ‘several individually distinct cows’ ?)

The coexistence of both *a and ⲉ |ɛ| / ⲏ |e| plurals for the same noun suggests that this vocalic opposition is not merely lexical, but rather encodes a morphosemantic distinction. Identifying the precise nature of this distinction, however, is difficult owing to the scarcity of the evidence: morphological plurals are rare in Coptic, and only a handful of nouns preserve both forms. Yet, on the basis of a very cursory overview of the forms, it seems to me that the ⲉ |ɛ| / ⲏ |e| plurals are primarily attested with highly individuated entities, such as human beings, deities, animals, settlements, and periods of time, whereas the **a*-plural exhibits a much wider range of uses and frequently occurs with more generic nouns, including those denoting objects and tools. This observation makes me wonder whether the **a*-plural represents the regular plural, encoding mere plurality of individual entities without further semantic specification, while the ⲉ |ɛ| / ⲏ |e| plurals may constitute a more semantically marked category, referring to collective groups of individuated entities or, perhaps, to sets construed at the level of kinds. A useful parallel may be found in the English plurals ‘fish’ vs ‘fishes’ (or ‘fruit’ vs ‘fruits’), where the unmarked plural denotes a generic plurality, while the marked plural is restricted to taxonomic or kind-level reference, referring to multiple species or varieties rather than merely multiple individuals. If such an analogy is appropriate, the Coptic ⲉ |ɛ| / ⲏ |e| plurals might likewise signal a higher degree of individuation or a different mode of reference than the more general **a*-plural. However, this suggestion remains highly speculative at present and would require a systematic reassessment of the available evidence before it could be properly evaluated.

18 For examples of nouns displaying both **a* and ⲉ |ɛ| or ⲏ |e| plurals, see Vergote 1969: 93–96.

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It is also important to observe that, even for those nouns that have only a single non-**a* plural, the stressed vowel in an open syllable is consistently ⲏ |e|, never ⲓ |i|. In other words, while there is evidence for a binary opposition between **a*-plurals and non-**a* plurals, there is no indication of a further morphological opposition between ⲏ |e| and ⲓ |i| within the non-**a* plural class. A few examples:

3	[S] ⲉⲛⲧⲏⲣ |ənté͡r| but also [S] ⲛⲧⲉⲉⲣ |nᵊté͡ːr| < **nV̲tÉrwV*	: ‘gods’
4	[S] ⲁⲙⲣⲏⲩ |amréw(ᵊ)| < **ʻVmrÉ(ỷ)wV* < ***ʻVmrÉ͡ː(ỷ)wV*	: ‘bakers’
5	[S] ⲁⲡⲏⲩⲉ |apéwə| < **(t)VpỶÉ(ỷ)wạt* < ***(t)VpỶÉ͡ː(ỷ)wạt*	: ‘heads’

II. Statives

Coptic statives also display a binary opposition between forms in **a* and forms with ⲉ |ɛ| / ⲏ |e|. While most verbs exhibit only one of the two patterns, a few attest both, suggesting that this variation is not merely paradigmatic but reflects a real morphological opposition.

This state of affairs had in fact already been noted by earlier scholars, who likewise suggested that the two vocalizations may reflect a morphological opposition. In particular, it has been proposed that statives in **a* may have carried a more qualitative value, whereas those in ⲉ |ɛ| / ⲏ |e| may originally have had a more passive or resultative meaning (see e.g. Lacau 1953: 12; see also Kilani 2025: 137–39). I am inclined to accept this analysis, which also appears to align well with what can be observed in the deverbal nouns (see below). It is important to stress that here as well the opposition appears to be binary, restricted to **a* versus ⲏ |e|, with no evidence for a corresponding ⲓ |i| counterpart. Examples of verbs with both **a*-statives and statives with ⲏ |e| are:

a. Statives displaying the pattern Cpt. CVCV́w < **CV³CV́ỷV*

1	[S] ϭⲁⲗⲏⲩ+ |kʲaléw(ᵊ)| < **qV³rÉỷạ*	: ‘dwells’,[19] ‘has come to dwell’ ?	2	[S] ϭⲁⲗⲱⲟⲩ+ |kʲalów(ᵊ)| < **qV³ráỷạ*	: ‘dwells’, ‘happens to be dwelling’ ?

b. Statives of reduplicated verbs displaying the pattern Cpt. (C¹V)C²VC³C²V́C³ < **C¹VC²VC³C²V́C³V*

1	[S] ⲗⲏⲗⲏⲛ+ |lᵊmlém(ᵊ)| < **kVnVmnÉmạ*	: ‘is twisted’, ‘has been twisted’ ?	2	[S] (ϭ)ⲗⲏⲗⲱⲛ+ |(kʲᵊ)lᵊmlóm(ᵊ)| < **kVnVmnámạ*	: ‘is twisted’, ‘happens to be twisted’ ?

19 I translate these statives using the 3rd person because this is the etymological origin of these forms, but of course in Coptic they could be used also with other persons.

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c. Statives of *2⁺-EtR* displaying the patterns CнC or CoCV < **CV́C̆V*

1	[S] ϣнк+ |šék| < **šÉq̆ạ*	: ‘is dug’, ‘has been dug’ ?	3	[S] ϣοκε+ |šókə| < **šáq̆ạ*	: ‘is dug’, ‘happens to be dug’
2	[S] пнс+ |pés| < **pÉšạ*	: ‘is cooked’, ‘has been cooked’ ?	4	[S] посε+ |pɔ́sə| < **pášạ*	: ‘is cooked’, ‘happens to be cooked’

d. Statives of *3-EtR* verbs with C^2 or C^3 = ʿ or *ʔ* < *r*, displaying the patterns CнC or CooCV < **CV́CCV*

1	[B] ϧнк+ |h₂ék| < **ẖÉʿqạ*	: ‘is scraped’, ‘has been scraped’ ?	4	[S] ϩοοκε+ |hɔ́ːkə| < **ẖáʿqạ*	: ‘is scraped’, ‘happens to be scraped’ ?
2	[B] ϫнр+ |čér| < **ḏÉrʿạ*	: ‘is scattered’, ‘has been scattered’ ?	5	[S] ϫοορε+ |čɔ́ːrə| < **ḏárʿạ*	: ‘is scattered’, ‘happens to be scattered’ ?
3	[S] тнϭ+ |tékʲ| < **dÉqʔạ* < ***dÉqrạ*	: ‘is fixed’, ‘has been fixed’ ?	6	[S] тοοϭε+ |tɔ́ːkʲə| < **dáqʔạ* < ***dáqrạ*	: ‘is fixed’, ‘happens to be fixed’ ?

e. Etymologically feminine statives with the pattern Cpt. *CVCV́wt* < **CV³CV́ẏtV*

1	[S] ϭаλнγт+ |kʲaléwt(ᵊ)| < **qV³rÉẏtV*	: ‘dwells’, ‘has come to dwell’ ?	2	[S] ϭаλοογт+ |kʲalɔ́wt(ᵊ)| < **qV³ráẏtV*	: ‘dwells’, ‘happens to be dwelling’ ?

III. Deverbal nouns

The case of deverbal nouns is different, because here the evidence for a morphological opposition underlying the different vocalizations comes from the semantic properties of these forms rather than from the existence of parallel morphological forms. In particular, Coptic has several deverbal nouns that can be interpreted either as quality/state nouns or as actant nouns. While preparing another study, I recently collected and published online[20] a curated corpus that includes 157 such nouns. These were selected because (a) their etymologies are generally unproblematic and widely accepted in previous scholarship

20 The database is available here: https://mkilani.github.io/ProtoCopticDeverbalDatabase/ For a preliminary discussion of the data from a presentation held in Basel in May 2026 see: https://github.com/MKilani/lcde-workshop/blob/main/presentations/vowels%201.3.pptx

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(e.g. Osing, Vycichl), (b) they are derived from strong 3-EtR verbal roots, and (c) these roots are already attested in the New Kingdom or earlier.

When the distribution of the vowels in these nouns is compared with their semantics, a clear correlation emerges. Excluding 22 nouns whose interpretation is inherently ambiguous, nouns with a stressed **a* vowel display predominantly active or qualitative semantics (62 out of 75), whereas nouns with ϵ |ɛ| / ⲏ |e| display predominantly passive/resultative or inchoative/reflexive semantics (49 out of 60). These results will be presented in detail in a paper currently in preparation. For present purposes, it is sufficient to note that they reveal a clear correlation between the type of vowel characterizing these deverbal nouns and a specific morphosemantic feature. This, in turn, implies that deverbal nouns displaying the same vocalic patterns and belonging to the same morphosemantic category (e.g. agent noun, state noun), but differing in the quality of their stressed vowel, may be regarded as morphological minimal pairs. Once this is taken into account, it becomes significant that, once again, when we consider forms with a stressed vowel in an open syllable, we observe only a binary opposition between forms whose vowel derives from **a* and forms with ⲏ |e|. Significantly, when compiling the corpus, I could not identify a single secure and unproblematic example of a noun with ⲓ |i| in an open syllable that could confidently be identified as a deverbal noun satisfying the criteria set out above: in other words, no opposition between ⲏ |e| and ⲓ |i| can be observed among deverbal nouns derived from strong 3-EtR roots. Once again, this supports the hypothesis that only two phonemic vowels need to be reconstructed for this position.

Examples from the corpus of deverbal nouns with a stressed vowel in an open syllable include:

a. Feminine quality-actant (patient/agent) nouns displaying the pattern Cpt. C(V)CCV́ < **CVCCV́ːv̆Vt*

1	[S] ⲙⲉⲥⲧⲏ |məsté|	: ‘hated person’	3	[S] ϩⲃⲥⲱ |həbsó|	: ‘garment’ < ‘the covering one’
	< **mVsḏÉv̆ạt*	< ***mVsḏÉːv̆ạt*		< **ḥabsáv̆ạt*	< ***ḥabsáːv̆ạt*
2	[S] ⲁⲗⲕⲏ |alké|	: ‘last day of the month’ < ‘the completed one’	4	[S] ⲉⲧⲡⲱ |ətpó|	: ‘burden’ < ‘the thing acting as a load’
	< **ʿVrqÉv̆ạt*	< ***ʿVrqÉːv̆ạt*		< **ʲVṯpáv̆ạt*	< ***ʲVṯpáːv̆ạt*

b. Feminine actant (patient/agent) nouns displaying the pattern Cpt. C(V)CV́C(V) < **CVCːˀV́ːˀCV(v̆)t*

1	[S] ⲟⲣⲏⲟⲉ |kʲᵊrékʲə|	: ‘dowry’ < ‘the established one’	2	[S] ⲙⲡⲱⲧ |mᵊpót(ᵃ)|	: part of loom < ‘the winding one’
	< **gVrÉgạ(v̆)t* < ***gVrːˀÉːˀgạ(v̆)t*			< **nVbádạt* < ***nVbáːˀdːˀạt*	

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c. Masculine actant (patient/agent) nouns displaying the pattern Cpt. C(V)CV́C(V) < **CVC:ˀV́:ˀCV(ẏ)*

1	[S] ⲡⲣⲏⲫ |pᵊréš|	: ‘mat’ < ‘the thing being spread’	4	[S] ⲁⲙⲟⲩⲛ |amún(ᵊ)|	: ‘Amoun’ < ‘the hiding one’
	< **pVrÉḫV(ẏ) < **pVr:ˀÉ:ˀḫV(ẏ)*			< **ʔVmánV(ẏ) < **ʔVm:ˀá:ˀnV(ẏ)*	
2	[S] ⲥⲧⲏⲙ |sᵊtém|	: ‘kohl’ < ‘the thing being applied (to the eyes)’	5	[B] ϭⲱϯ |kʲᵊjótɨ|	: ‘tip of scorpion‘s tail’ < ‘the stabbing one’
	< **sVdÉmV(ẏ) < **sVd:ˀÉ:ˀmV(ẏ)*			< **ḏV³ádV(ẏ) < **ḏV³:ˀá:ˀdV(ẏ)*	
3	[S] ϣⲉⲛⲏⲧ |šənét(ᵊ)|	: ‘shark (?)’ < ‘the reviled one’	6	[S] ⲥⲁⲛⲟⲩⲑ |sanútʰ(ᵊ)|	: ‘fearful’ < ‘the fearing one’
	< **šVnÉṯV(ẏ) < **šVn:ˀÉ:ˀṯV(ẏ)*			< **sVnáḏV(ẏ) < **sVn:ˀá:ˀḏV(ẏ)*	

5.2.3 Morphological forms exclusively associated with either ⲏ |e| or ⲓ |i|

The examples discussed above show that, while morphological oppositions between **a* and ⲏ |e| are well attested in Coptic, no comparable opposition can be identified between ⲏ |e| and ⲓ |i|. This strongly suggests that ⲏ |e| and ⲓ |i| can be understood as mere allophonic realizations of a single phonemic vowel, which in turn contrasted with **a*.

From this perspective, it is also worth noting that the vowels ⲏ |e| and ⲓ |i| can each be associated exclusively with particular morphological forms. In these cases, the vowels do not participate in any morphological opposition with other vowels. Rather, the fact that each morphological form consistently displays either ⲏ |e| or ⲓ |i| can once again be interpreted as reflecting the complementary distribution of two allophones of a single phonemic vowel, with the choice of allophone conditioned by (the phonological features of) the morphological form itself. The main examples are as follows:

IV. Forms with ⲏ |e|

a. In a reduplicated form ⲛⲏ |e:| in nouns with C^2 or C^3 = *ʔ* < *r*/ʿ/ʾ and displaying a pattern CⲛⲏC(ⲉ) < **CVCCVẏ* (masculine) or **CVCCV(ẏ)t* (feminine); in these forms ⲛⲏ |e:| can alternate with ⲏ |e|, ⲉ |ɛ| or even ⲁⲓ |aj|

	Masculine	
1	[S] ⲟⲩⲛⲏⲃ |wé:b| < **wÉʿbVẏ*	: ‘priest’
2	[S] ⲙⲛⲏϣⲉ |mé:šə| < **mÉšʿVẏ*	: ‘multitude’, ‘crowd’, ‘troop’
	Feminine	
3	[S] ⲧⲛⲏⲃⲉ |té:bə|, ⲧⲛⲃⲉ |tébə|, ⲧⲁⲓⲃⲉ |tájbə|, ⲧⲉⲃⲓ |tébi| < **ḏÉbʲV(ẏ)t*	: ‘chest’, ‘coffin’, ‘pouch’

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b. Pre-pronominal form of the infinitive of *3-EtR* verbs displaying the pattern C(V)CCV́t= < **CVCCV́(ỷ)t=*

1	[S] ⲥⲙⲥⲏⲧ= |šˢmšét=| (Abs. [S] ⲥⲙⲥⲉ) < **šEmsÉ(ỷ)t=*	: ‘to serve’, ‘to worship’
2	[B] ⲥⲁϩⲛⲏⲧ= |sahnét=| (Abs. [B] ⲥⲁϩⲛⲓ) < **sEḥnÉ(ỷ)t=*	: ‘to provide’, ‘to supply’

V. Forms with ⲓ |i|

a. Infinitives of *2⁺-EtR* verbs with the pattern CV́CV < **CV́C̆Vt*

1	[S] ⲡⲓⲥⲉ |písə| < **pÉšE̯t*	: ‘to boil, cook’, ‘to be cooked’
2	[S] ⲟⲩⲉⲓⲛⲉ |wínə| < **wÉn̆E̯t*	: ‘to pass’
3	[S] ⲥⲓⲕⲉ |síkə| < **sÉk̆E̯t*	: ‘to grind’

b. Pre-pronominal form of the infinitive of *2⁺-EtR* verbs displaying the pattern Cpt. (CV)CV́t= < **CVC̆V́(ỷ)t=*

1	[S] ⲙⲉⲣⲓⲧ= |mᵊrít=| (Abs. [S] ⲙⲉ) < **mVr̆É(ỷ)t=*	: ‘to love’
2	[S] ⲛⲓⲧ= |nít=| (Abs. [F] ⲛⲓ ?) < **(i̯V)n̆É(ỷ)t=*	: ‘to bring’, ‘to bear’
3	[S] ⲫⲓⲧ= |fít=| (Abs. [S] ⲫⲓ ?) < **fV̆ʲÉ(ỷ)t=*	: ‘to bear’, ‘to carry’, ‘to take’

c. Pre-pronominal form of imperatives of *2⁺-EtR* verbs with **Cᴵ = *i̯* and extended with the suffix **-Vt=*, characterized by the pattern Cpt. CVCV́t= < **CVC̆V́t=*

1	[B] ⲁⲣⲓⲧ= |arít=| (inf. [S] ⲓⲣⲓ) < **i̯Vr̆Ét=*	: ‘act!’, ‘make!’, ‘do!’
2	[B] ⲁⲛⲓⲧ= |anít=| (inf. [B] ⲓⲛⲓ) < **i̯Vn̆Ét=*	: ‘bring!’ ‘bear!’

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d. *Nisba* nouns characterized by a pattern C(V)CV́t < **CVC̆V́ẏtVj* and in turn derived from *2⁺-EtR *CVC̆V́:ʔẏVt* feminine nouns[21]

	Masculine	
1	[S] ⲙⲉⲓⲧ |mᵊhít(ᵊ)|	: 'north'
	< **mVḫ̆É:ʔẏtE̱j*, derived from **mVḫ̆É:ʔẏạt* = [L] ⲙⲉⲏ |mᵊhé|	: 'northwind'
2	[S] ϩⲟⲩⲉⲓⲧ |hᵊwít(ᵊ)|	: 'first' (masculine)
	< **ḥV̆É:ʔẏtE̱j*, derived from **ḥV̆É:ʔẏạt* = [S] ⲉϩⲏ |ᵊhé|	: 'beginning', 'front part'
	Feminine	
3	[S] ϩⲟⲩⲉⲓⲧⲉ |hᵊwítə|	: 'first' (feminine)
	< **ḥV̆É:ʔẏtE̱(j)t*, derived from **ḥV̆É:ʔẏạt* = [S] ⲉϩⲏ |ᵊhé|	: 'beginning', 'front part'

d. To this list, we can also add the forms of the numeral 'nine' (sg.) / 'ninety' (pl.) (from Vycichl 1984: 248).

sg.m.	sg.f.	pl. ('ninety')
[S] ϥⲓⲧ |pᵊsítᵊ|	[S] ϥⲓⲧⲉ |pᵊsítə|	[S] ⲡⲥⲧⲁⲓⲟⲩ |pᵊsᵊtájwᵊ|
< **pVsÉ̱dE(ẏ)*	< **pVsÉ̱dE(ẏ)t*	< **pVsE̱dÉẏwV*
< ***pVsːʔÉːʔ̱dE(ẏ)*	< ***pVsːʔÉːʔ̱dE(ẏ)t*	< ***pVsːʔEːʔ̱dÉẏwV*

The singular forms of 'nine' are stressed on the second syllable, whereas the plural form is stressed on the third. This makes it possible to reconstruct both vowels, and their Coptic reflexes point to **E* in both cases. The post-tonic vowel in the feminine form can be reconstructed by analogy with the masculine.

5.2.4 Morphological contrasts: discussion

Since ⲏ |e| and ⲓ |i| never occur in opposition to one another or together in opposition to **a*, there is no compelling reason to assume that ⲏ |e| and ⲓ |i| reflect two distinct phonemic proto-vowels. Rather, the distribution just described can be straightforwardly interpreted as reflecting two allophonic realizations of a single phonemic vowel, conditioned by the phonological environment of each morphological form. The data presented above, in fact, reveal two specific patterns that strongly support such an interpretation.

The first concerns the contrast between the cases IV.b and V.b. In the pre-pronominal infinitive in *-t*, **E* appears to surface as ⲏ |e| when the verb has three strong consonants (IV.b), but as ⲓ |i| when the verb has only two strong consonants (V.b). While it is

21 I interpret these nouns as **CVC̆V́:ʔẏVt*, although in principle they could also be analysed as **CVCV́:ʔXVt*. Under the latter interpretation, the pattern underlying the nisba would be **CVCV́XtVj*. I prefer the reconstruction **CVC̆V́:ʔẏVt*, since positing a consonant cluster *C̆* rather than a plain consonant may help account for the appearance of ⲟⲩ /w/ in ϩⲟⲩⲉⲓⲧⲉ /hᵊwítə/, which would then derive from **-ʔ̆* rather than from plain **-ʔ-*. This evidence, however, is admittedly weak and remains speculative.

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theoretically possible that verbs with different consonantal structures contained different phonemic vowels, a more straightforward and economical explanation is that both forms shared a same single phonemic vowel whose surface realization varied according to the phonological context created by the presence of two vs three strong consonants in the root of the verbs.

The second pattern is provided by the *nisba* nouns discussed in V.d. Here, the stressed vowel of the derived *nisba* remains in the same position as in the corresponding base noun, since the position of the stress does not change. Yet, their surface realizations differ, being ⲏ |e| in the base noun but ⲓ |i| in the derived *nisba*. There is no reason to assume that the phonemic vowel of the base noun was replaced by a different phonemic vowel in the derived form. Instead, the most plausible interpretation is that both base nouns and their *nisba* derivatives contained the same underlying phonemic vowel, whose realization as ⲏ |e| or ⲓ |i| was determined by the respective phonological environments.

If the vowels ⲏ |e| and ⲓ |i| represent allophones of the same phoneme, then the crucial question is which factors condition their surface realization. As a starting point, we can rule out the strong consonants that survive in Coptic, since the patterns under discussion are general morphological phenomena and do not appear to be restricted to roots with specific strong consonants. Instead, it is more plausible that etymological weak consonants, or consonants that were weakened or lost before Coptic, played a role. For example, the presence of ⲏⲏ |e:| in IV.a clearly correlates with the presence of a weak consonant following the stressed vowel. Similarly, the distinction between verbs with three strong consonants and those with two in IV.b and V.b can be reformulated as verbs without weak consonants, that is, *3-EtR* verbs, versus verbs with one weak consonant, assuming that *2⁺-EtR* roots originally contained (at least) one weak consonant (see Kilani 2025: 142–43). However, the presence or absence of weak consonants alone does not fully account for the distribution of ⲏ |e| and ⲓ |i|. This is evident in verbs with a CⲏC stative that simultaneously exhibit a CⲓCⲉ infinitive. If the ⲏ |e| versus ⲓ |i| alternation depended solely on the presence of weak consonants in the root, such a coexistence of patterns within the same roots would not be possible.

Another possibility is that the realization of these stressed vowels was conditioned by nearby unstressed vowels through processes such as vowel harmony or metaphony. The challenge in this case is that unstressed vowels are notoriously difficult to reconstruct in Proto-Coptic, as in Coptic they are regularly reduced and often merge into a single indistinct vowel. Nevertheless, differently stressed secondary forms can occasionally provide evidence for the nature of unstressed vowels. For example, the first unstressed vowel of IV.b can be reconstructed as **E* on the basis of the absolute form of the infinitives, in which this vowel is stressed.

Indeed, analogy can also be deployed as a reconstructive tool, though it must be applied with caution. The CⲓCⲉ < **CV́C̆-Vt* infinitives illustrate this approach. In these forms, the posttonic vowel is never attested in a stressed context in Coptic, and therefore it cannot be directly reconstructed. However, some of these verbs also exhibit a second type of infinitive characterized by a similar pattern CⲉC(ⲉ) < **CV́C̆y̆-Vt*. In this pattern

!!! Uncorrected proof (type-set, pre-print)!!!

both the first and second vowels can be stressed (in the absolute and prepronominal states respectively), allowing both vowels to be safely reconstructed; in both cases they appear as **E*. Since the *-Vt* suffix is likely identical in both infinitive types, and since it can be reconstructed as **-Et* in the second pattern, analogy supports reconstructing it as **-Ḥt* in the first pattern as well, which can thus be vocalized as **CÉC̆-Ḥt*. This is in fact the most widely accepted reconstruction for this infinitive (see e.g. Loprieno 1995: 89; Vycichl 1984: passim).

Finally, comparison with other Afroasiatic languages, especially Semitic, can also provide guidance for proposing tentative vocalizations for selected morphemes and morphological endings. For instance, the 3sg.m stative ending is generally reconstructed as **-a* in Semitic (Huehnergard 2019: 63). Given the striking similarity with Semitic stative endings, it seems indeed reasonable to reconstruct the Egyptian form as **-ạ*, as has been suggested by previous scholars as well (see e.g. Loprieno 1995: 78; Allen 2020: 134). The less common *-tV* ending of the stative has also been reconstructed as **-tạ* (Allen 2020: 134), but the evidence is admittedly weaker and has thus not been adopted here.

The same reasoning applies to the Egyptian non-nisba feminine suffix, which is reconstructed here as **-ạt*, since this is the form unanimously reconstructed for the corresponding suffix in Proto-Semitic (Huehnergard 2019: 58). This reconstruction is further supported by a few rare Coptic forms derived from original dual forms in which the vowel of the feminine suffix was stressed and appears as **a*, such as [S] ⲥⲧⲱⲧⲉ |sˁtótə| < **sVt̲ʾát-Vw̆* : 'two arouras' or [S] ⲥⲡⲟⲧⲩ |sˁpɔ́tw(ˁ)| < **sVpát-(V)w̆=Vw* : 'their (two) lips' (cf. Loprieno 1995: 60).

As for the nisba endings, the Semitic languages attest two sets of forms: one characterized by the vowel **a* and another by the vowel **i* ; for example, Akkadian m. *-āy-* , f. *-āyīt-* versus m. *-ī-*, f. *-īt-* (see Butts 2010: 81–83 for an overview of Semitic forms). The Egyptian endings are usually reconstructed as **-Ḥ(j)* and **-Ḥt* (Loprieno 1995: 58), thus aligning more closely with the Semitic forms containing **i* rather than **a*.

Now, when we take into account all the unstressed posttonic vowels that can be reconstructed for the grammatical categories discussed above, an interesting pattern emerges. Statives (II), feminine plurals (I.5) and non-nisba feminine nouns (III.a, III.b), that is, all forms in which the posttonic vowel can be reconstructed as **ạ*, are consistently characterized by a stressed vowel ⲏ |e|. In contrast, the nisba C(V)CV́t < **CVC̆V́w̆tVj* (V.d), the infinitives CV́CV < **CV́C̆Vt* (V.a), and the number 'nine' (V.e), namely all the forms that can reasonably be reconstructed with a posttonic vowel **E*, are characterized by a stressed vowel ⲓ |i|. If these reconstructions are correct, this observation suggests a rather simple and neat set of rules, namely:

- *R.1* In penultimate *CV* or *CVw̆* syllable, **É* becomes ⲓ |í| before unstressed **E*
- *R.2* In penultimate *CV* or *CVw̆* syllable, **É* becomes ⲏ |é| before unstressed **a*

To which we can add the following rules, on the basis of the previous considerations:

- *R.3 *É* becomes ⲓ |í| in the environment *-C̆ÉC=*
- *R.4 *É* becomes ⲏ |é| in the environment *-CÉC=*

!!! Uncorrected proof (type-set, pre-print)!!!

- *R.5* ***É*** becomes ⲏ |é| or ⲏⲏ |é:| (as well as ⲉ |ɛ| or ⲁⲓ |aj|[22]) in forms *CÉCV* < *C*[1]*VC*[2]*C*[3]*V* where C[2] or C[3] = *ʔ* < *r*/*ʿ*/*ɣ*

It appears that these rules are sufficient to account for *all* the cases discussed above. Moreover, they also provide an elegant solution for the particularly puzzling case of the nisba C(V)CV́t < ****CVC̆V́ytVj*** (V.d), whose vowel seems to change between the basic noun and its derived nisba. Simply put, in the basic form of the noun, the stressed vowel ****É*** surfaces as ⲏ |é| because it is directly followed by the ****ạ*** of the feminine ending ****-ạt***. In contrast, the addition of the nisba suffix triggers the regular syncope of the ****ạ***, so that the posttonic vowel is now that of the nisba suffix, which can be reconstructed as ****E***. This, in turn, conditions the stressed ****É*** to be realized as ⲓ |í|.

Finally, the data above reveal another interesting pattern. In I, II.c, and II.d, ⲏ |e| appears to contrast with ⲟ |ɔ| and its variants. At first glance, this might seem unexpected, since ⲏ |e| is typically associated with open syllables, while ⲟ |ɔ| occurs in closed syllables. However, a closer examination of the contexts in which this contrast occurs shows that these stressed vowels are always followed either by a *C̆* consonant or by a two-consonant cluster in which at least one consonant was etymologically *ẏ*, *w*, or *ʔ* < *r*/*ʿ*/*ɣ*.

This observation is important for two reasons. First, it supports the idea that *C̆* consonants behave differently from regular consonants, acting more like consonant clusters than single consonants. Second, it suggests that the feature characterizing *C̆* consonants was akin to the presence of a weak (*ẏ*, *w*) or weakened (*ʔ* < *r*/*ʿ*/*ɣ*) consonant. At the same time, it indicates that this feature could assume a syllabic role after stressed ****a***, but was neutralized or assimilated when it followed a stressed ****E***.

All of these observations can be summarized in two additional rules:

- R.6 ***CÉC̆V***, ***CÉC*[*a*]*C*[*b*]*V*** where *C*[*a*] or *C*[*b*] = *ẏ, w, ʔ* < *r*/*ʿ*/*ɣ* become CⲏC(V)
- R.7 ***CáC̆V***, ***CáC*[*a*]*C*[*b*]*V*** where *C*[*a*] or *C*[*b*] = *ẏ, w, ʔ* < *r*/*ʿ*/*ɣ* become CⲟC(V)

6 Summary and conclusions

The evidence discussed here shows, I think, that it is sufficient to assume only two Proto-Coptic phonemic vowels in order to account for the Coptic data. At the same time, as noted at the beginning, any reconstruction is necessarily constrained by what is actually preserved in the available evidence. In this sense, the model proposed here can also be understood in explicitly minimalist terms: a *minimum* of two vowels is required to explain the Coptic evidence.

However, on the basis of Egyptian–Coptic evidence alone, it cannot be established whether the two-vowel system proposed here can be projected back to Pre-Proto-Coptic or even to Proto-Afroasiatic, or whether it instead reflects the simplification of an earlier system with a richer vocalic inventory. Addressing this question would require a broader comparative Afroasiatic perspective and therefore lies beyond the main scope of the

22 The factors that occasionally cause ****E*** to surface as ⲏⲏ |e:|, ⲉ |ɛ|, or ⲁⲓ |aj| in these contexts remain unclear.

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present paper. Still, it is worth observing here that, as noted above, even a cursory look at the other major branches of the family reveals a picture that is far from uniform. On the one hand, the proto-languages of several Chadic branches are often reconstructed with essentially binary vocalic oppositions that closely resemble the system proposed here for Proto-Coptic. On the other hand, the proto-languages of various Cushitic branches are commonly reconstructed with five or more proto-vowels. Between these poles stand Proto-Semitic, with its three short and three long vowels, and Proto-Berber, which has been reconstructed in different ways, most recently as a mixed system with three or four long vowels but only two short ones. Overall, the range of vowel inventories reconstructed across Afroasiatic is so broad and diverse that it offers little insight into the historical origin or broader status of the binary system proposed here for Proto-Coptic.

At the same time, vowels themselves may not be the only piece of the puzzle. Pre-Coptic Egyptian has long been known to have been characterized by widespread palatalization and probably also labialization of consonants (Peust 1999: 107–15, 119–25). Moreover, palatalization and labialization are prominent features across most branches of the Afroasiatic family, with Semitic as exception (apart from Ethio-Semitic, see e.g. Gragg 1997: 244). In Chadic, moreover, the palatalization and labialization of consonants often interact with binary vocalic systems, giving rise to additional, contextually conditioned allophonic vowels, most notably *i* and *u*, at the surface level (Wolff 2008: 47–49, 2022). Comparable interactions between palatalized or labialized consonants and a reduced set of phonemic vowels have also been proposed for Berber (see Kossmann 2020: 34–37), and similar effects can even be observed in some Berber-influenced varieties of North African Arabic, where original /u/ may be realized as *ə* + *palatalization of a neighbouring consonant* – cf. e.g. Moroccan Arabic *ḫʷəbz* < *ḫubz* 'bread' (van Putten, personal communication). This raises the question of whether the binary vocalic system reconstructed in this paper represents only part of a more complex, and possibly earlier, phonological reality, in which the realization of a minimal set of vowels was closely intertwined with patterns of palatalization and labialization in adjacent consonants. In fact, cross-linguistic and typological observations suggest that reduced vocalic systems often tend to exhibit complex interactions with glides and palatal consonants (see the discussion in Wolff 2025: 253–55 in relation to Chadic). Clarifying this issue will require further research.

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